Event Management System — Requirements

(10 Functional, 7 Non-Functional)

1. Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow users to create an account using their name, email, and password. The user's email must be verified before the account is activated.	High	Pass: The account is created but remains inactive and cannot be used to log in until the user clicks the verification link sent by email. After verification, login succeeds. Fail: The account becomes active without email verification.	Ensures that user identities are verified before allowing login or role-based access.
FR-002	Functional	The system shall allow users to log in using their email and password and reset a forgotten password through an emailed reset link.	High	Pass: Valid credentials allow login. Selecting "Forgot Password" sends a time-limited reset link that successfully changes the password, making the old password invalid. Fail: Invalid credentials are accepted, or the reset link does not expire or fails to update the password.	Provides essential authentication and password recovery functionality.
FR-00	Functional	The system shall assign each user a role (Admin, Organizer, or	High	Pass: Each role can perform only its	Maintains proper access control and

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3		Participant) and restrict access to features based on that role. Users shall be able to update their own profile information, while admins shall be able to view, suspend, or delete user accounts.		permitted actions; profile changes are saved correctly; suspended or deleted accounts are immediately prevented from logging in. Fail: A role can perform an unauthorized action, or a suspended account can still log in.	allows administrators to manage user accounts.
FR-004	Functional	The organizer shall be able to create an event by providing its name, description, date, time, venue, category, and capacity. The organizer shall also be able to modify the event details after creation.	High	Pass: All required event details are stored correctly when an event is created, and subsequent changes are saved and displayed immediately. Fail: An event is created without a required field, or an update is not saved.	Provides the core functionality required for organizers to create and manage events.
FR-005	Functional	The organizer shall be able to delete or cancel an event. If an event that has registered participants is updated or cancelled, the system shall notify all affected participants.	High	Pass: A cancelled event no longer accepts registrations, and all registered participants receive a notification within 5 minutes. Fail: The event continues accepting registrations after cancellation, or any registered participant fails to receive the notification.	Ensures participants are promptly informed about changes that may affect their plans.
FR-	Function	Users shall be able to view	High	Pass: The event list	Makes it easier for

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006	al	available events and search by event name or filter events based on category, date, or venue. The admin shall be able to manage event categories.		displays all upcoming, non-cancelled events. Search and filter results contain only relevant events, and category changes made by the admin are reflected throughout the system. Fail: Events are incorrectly included or excluded, or category changes are not updated properly.	users to discover relevant events and ensures accurate search and filtering.
FR-007	Functional	A logged-in user shall be able to register for an event. The system shall store the user's registration details, including the user, event, timestamp, and status, while preventing duplicate registrations for the same event.	High	Pass: A valid registration creates exactly one correct record for the user-event combination, while a second registration attempt is rejected. Fail: Duplicate registrations are created or a valid registration is not recorded.	Ensures reliable event registration and maintains data integrity.
FR-008	Functional	The system shall verify the remaining event capacity before confirming a registration. Registrations shall be rejected when an event is full, and concurrent registration requests shall be handled without exceeding the available capacity.	High	Pass: Registration for a full event is rejected. For 100 simultaneous requests to a 1-seat event, exactly one request receives confirmation. Fail: A full event accepts a registration or multiple users are confirmed for the same final seat.	Prevents overbooking, which is one of the most critical failure scenarios for the platform.

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FR-009	Functional	When an event reaches full capacity, users shall be given the option to join a waitlist. If a registered participant cancels, the system shall automatically offer the available seat to the next person on the waitlist. Users shall also be able to cancel their own registrations, and the system shall send registration, cancellation, and event-reminder notifications.	Medium	Pass: Waitlist entries are recorded in the correct order; when a seat becomes available, the first waitlisted user is automatically promoted and notified; self-cancellation updates the registration status and seat count immediately; required confirmation and reminder emails are sent within their specified time windows. Fail: An available seat remains unused despite a waitlisted user, or a required notification is not sent.	Maximizes registrations after cancellations and keeps users informed throughout the registration process.
FR-010	Functional	The system shall display the number of registered participants and available seats for each event. Organizers shall be able to view event-specific registration details and export participant lists in CSV or PDF format. Admins shall be able to view overall platform statistics, including total events, users, and registrations.	Medium	Pass: Displayed and exported counts accurately reflect current active registrations, and administrator statistics match the live database totals. Fail: Any displayed or exported information is outdated or inaccurate.	Provides organizers and administrators with the information needed to manage, monitor, and evaluate the platform.

2. Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The system shall load event list and search/filter pages within 2 seconds, process a registration (capacity check, record creation, confirmation) within 3 seconds, and generate CSV or PDF participant exports of up to 5,000 registrations within 10 seconds, each for at least 95% of requests under normal load (up to 500 concurrent users).	High	Pass: In a load test with 500 concurrent users, the 95th percentile is 2 seconds or less for event list, search and filter, and 3 seconds or less for registration; a 5,000-record export completes within 10 seconds. Fail: Any of these targets is exceeded.	Fast browsing and registration keep users engaged, and organizers need timely attendee data.
NFR-002	Scalability and Concurrency	The system shall support at least 10,000 users, 1,000 active events and 100,000 registrations without degradation, and shall execute concurrent operations on shared data (seat count, waitlist order, duplicate checks) atomically so that no overbooking, duplicate registration or lost update occurs.	High	Pass: With the dataset above loaded, NFR-001 targets still hold; 100 simultaneous requests for a 1-seat event confirm exactly one; 100 simultaneous duplicate requests from one user create exactly one record; waitlist order matches request order. Fail: Any overbooking, duplicate or out-of-order waitlist entry occurs, or targets are missed at this volume.	Overbooking is one of the most critical failure scenarios, and the platform must grow without a rebuild.
NFR-003	Security (Data Protection)	The system shall store passwords only as salted hashes (bcrypt, scrypt or Argon2), never in plain text or logs; verification and reset links shall be single-use, random and time-limited; all traffic shall use	High	Pass: Database and log review show no plain-text passwords; a used or expired link is rejected; HTTP redirects to HTTPS and a TLS scan finds no version below 1.2; database storage is encrypted. Fail: Any plain-text password, reusable	Protects credentials and personal data against interception and database compromise.

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		TLS 1.2 or higher; and sensitive data at rest shall be encrypted.		link, unencrypted connection or unencrypted stored data is found.	
NFR-004	Security (Access Control)	Role-based access control shall be enforced on the server for every request, not only in the interface. Sessions shall expire after 30 minutes of inactivity, accounts shall lock after 5 consecutive failed logins, the system shall be protected against OWASP Top 10 attacks, and administrative actions (role changes, suspension, deletion, event cancellation) shall be recorded in an audit log kept for 12 months.	High	Pass: Direct API calls by a role without permission return 403; an idle session is rejected after 30 minutes; the account locks after 5 failed attempts; a security scan reports no high or critical findings; each listed action creates a log entry with user, action and timestamp. Fail: Any unauthorized action succeeds, a high-severity vulnerability is found, or an action is not logged.	Server-side enforcement is the only reliable access control and supports immediate suspension of accounts.
NFR-005	Availability and Recovery	The system shall be available at least 99.5% of the time each month, excluding announced maintenance of no more than 4 hours per month. Data shall be backed up at least daily and be restorable with a recovery point objective of 24 hours or less and a recovery time objective of 4 hours or less.	High	Pass: Monthly monitoring shows uptime of 99.5% or higher; a restore drill completes within 4 hours and loses no more than 24 hours of data. Fail: Uptime falls below 99.5%, backups are missing or corrupt, or restoration exceeds the targets.	Users and organizers must be able to rely on the platform, especially close to event dates, and data must survive failures.
NFR-006	Usability and Accessibilit	A first-time participant shall be able to find an event and register within 3	Medium	Pass: At least 90% of first-time users complete the task within 3 minutes in testing; an	Low friction improves registration rates

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	y	minutes without training, with clear error messages for cases such as a full event or duplicate registration. The interface shall meet WCAG 2.1 Level AA, be responsive from 320 px upward, and work on the latest two versions of Chrome, Firefox, Safari and Edge.		accessibility audit shows no Level AA failures; all key flows work on the listed browsers and on mobile screens. Fail: Fewer than 90% succeed, Level AA failures remain, or a key flow breaks on a supported browser or device.	and ensures everyone, including users with disabilities and on mobile, can use the platform.
NFR-007	Maintainability	Application errors shall be logged centrally with timestamp and severity, and the codebase shall have automated test coverage of at least 80% for core business logic (registration, capacity check, waitlist promotion, notifications).	Medium	Pass: A forced error appears in the central log with timestamp and severity; the coverage report shows 80% or more for the core modules. Fail: Errors are missing from the log, or coverage is below 80%.	Makes problems quick to diagnose and future changes safe to release.

3. Traceability Summary

Functional Requirement	Supporting Non-Functional Requirements
FR-001 Account creation	NFR-003
FR-002 Login and password reset	NFR-003, NFR-004
FR-003 Roles and user management	NFR-003, NFR-004
FR-004 Create and modify events	NFR-005
FR-005 Cancel events and notify	NFR-004, NFR-007
FR-006 Browse, search, filter	NFR-001, NFR-002, NFR-006
FR-007 Register for events	NFR-001, NFR-002, NFR-005, NFR-006
FR-008 Capacity check	NFR-001, NFR-002, NFR-006, NFR-007

Functional Requirement	Supporting Non-Functional Requirements
FR-009 Waitlist and notifications	NFR-002, NFR-007
FR-010 Counts, export, statistics	NFR-001, NFR-002, NFR-005